



MTH603 - Numerical Analysis



MTH603 - Numerical Analysis

(Mid-Term) 2021 GRAND Quiz 90 TO 100% Correct Solve (Alhamdulillah)

BY Virtual Assistance Sonu Ilyas Mughal



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03063548328 Sonu Ilyas Mughal



Question # 1 of 30 (Start time: 04:08:34 PM, 28 June 2021)

Eigenvalues of a symmetric matrix are all _____ .

Select the correct option

☐	positive
☒	real
☐	complex
☐	zero

Click to Save Answer &





MTH603:Grand Quiz

Question # 2 of 30 (Start time: 04:09:11 PM, 28 June 2021)

Jacobi's Method is a/an.....

Select the correct option

- | | |
|----------------------------------|------------------|
| ☒ | Iterative method |
| ☐ | Direct method |

Click





MTH603:Grand Quiz

Question # 3 of 30 (Start time: 04:09:32 PM, 28 June 2021)

In method, a system is reduced to an equivalent diagonal form using elementary transformations.

Select the correct option

☐	Gauss-Seidel
☐	Jacobi's
☐	Relaxation
☒	Gaussian elimination





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Question # 4 of 30 (Start time: 04:10:01 PM, 28 June 2021)

Which method requires a derivative of the solution?

Select the correct option

☐	Regula-Falsi method
☒	Newton-Raphson method
☐	Bisection method
☐	Muller method





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Question # 5 of 30 (Start time: 04:10:30 PM, 28 June 2021)

The first row of the augmented matrix of the system of linear equations is:

$$2x+z=4$$

$$x-y+z=-3$$

$$-y+z=-5$$

Select the correct option

- | | |
|----------------------------------|----------------|
| ☐ | 1, 0, 0 and 4 |
| ☐ | 2, 0, 1 and -4 |
| ☐ | 0, 2, 1 and 4 |
| ☒ | 2, 0, 1 and 4 |





Question # 6 of 30 (Start time: 04:10:56 PM, 28 June 2021)

For a system of linear equations, the corresponding coefficient matrix has the value of determinant; $|A| = -3$, then which of the following is true?

Select the correct option

- | | |
|----------------------------------|--|
| ☒ | The system has unique solution |
| ☐ | The system has finite multiple solutions |
| ☐ | The system has no solution |
| ☐ | The system has infinite may solutions |





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Question # 7 of 30 (Start time: 04:11:19 PM, 28 June 2021)

If $n \times n$ matrices A and B are similar, then they have the _____ eigenvalues (with the same multiplicities).

Select the correct option

☐	different
☒	same

Click to S





Question # 8 of 30 (Start time: 04:11:46 PM, 28 June 2021)

While using Gaussian Elimination method, the following augmented matrix

$$[A|I] = \begin{bmatrix} 1 & \frac{3}{4} & 0 & \frac{11}{40} & \frac{1}{5} & -\frac{1}{40} \\ 0 & 1 & 0 & -\frac{3}{2} & 0 & 1 \\ 0 & 0 & 1 & \frac{11}{40} & -\frac{1}{5} & -\frac{1}{10} \end{bmatrix}$$

will reduce in identity matrix by performing

Select the correct option



- | | |
|----------------------------------|------------------------|
| ☐ | $R_2 - \frac{3}{4}R_1$ |
| ☐ | $R_2 - \frac{3}{4}R_1$ |
| ☒ | $R_1 - \frac{3}{4}R_2$ |
| ☐ | $R_1 - \frac{4}{3}R_2$ |

Click to Save Answer & Mark





Question # 9 of 30 (Start time: 04:12:13 PM, 28 June 2021)

The basic idea of relaxation method is to reduce the largest residual to

Select the correct option

☒	Zero
☐	One
☐	Two
☐	None of the given choices





Question # 10 of 30 (Start time: 04:12:35 PM, 28 June 2021)

While using Jacobi method for the matrix

$$A = \begin{bmatrix} 1 & 1/4 & 1/3 \\ 1/4 & 1/3 & 1/2 \\ 1/3 & 1/2 & 1/5 \end{bmatrix}$$

and 'theta $\theta=0.7191$ ', the orthogonal matrix S_1 will be given as

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | $S_1 = \begin{bmatrix} \cos 0.7191 & 0 & 0 \\ 0 & -\sin 0.7191 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ |
| ☐ | $S_1 = \begin{bmatrix} \cos 0.7191 & 0 & -\sin 0.7191 \\ 0 & 1 & 0 \\ \sin 0.7191 & 0 & \cos 0.7191 \end{bmatrix}$ |
| ☐ | $S_1 = \begin{bmatrix} \cos 0.7191 & -\sin 0.7191 & 0 \\ \sin 0.7191 & \cos 0.7191 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ |
| ☒ | $S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 0.7191 & -\sin 0.7191 \\ 0 & \sin 0.7191 & \cos 0.7191 \end{bmatrix}$ |





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Question # 11 of 30 (Start time: 04:13:04 PM, 28 June 2021)

The number system that has a base 16 is called...system.

Select the correct option

☐	Octal
☒	Hexadecimal
☐	Decimal
☐	Binary





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Question # 12 of 30 (Start time: 04:13:34 PM, 28 June 2021)

The Gauss-Seidel method is applicable to strictly diagonally dominant or symmetric positive definite matrices A.

Select the correct option

☒	TRUE
☐	FALSE





Question # 13 of 30 (Start time: 04:14:00 PM, 28 June 2021)

The eigenvectors of a square matrix are the non-zero vectors that, after being multiplied by the matrix, remain to the original vector.

Select the correct option

☐	Diagonal
☐	None of the given choices
☒	Parallel
☐	Perpendicular

Click to Save Answer & Move to Next Question





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Question # 14 of 30 (Start time: 04:14:28 PM, 28 June 2021)

Numerical methods for finding the solution of the system of equations are classified as direct and methods

Select the correct option

☐	Jacobi
☒	Iterative
☐	None of the given choices
☐	Indirect





Question # 15 of 30 (Start time: 04:14:54 PM, 28 June 2021)

Tc

While using power method, from the resultant normalize vector

$$u^{(3)} = 3.17890 \begin{pmatrix} 1.0 \\ 1.231926 \\ 1.421389 \end{pmatrix}$$

we have the largest eigen value and the corresponding eigenvector as

Select the correct option

[Reload Math](#)

- | | |
|----------------------------------|--|
| ☐ | $\lambda = 3.17890, (X) = \begin{pmatrix} 1.231926 \\ 1.0 \\ 1.421389 \end{pmatrix}$ |
| ☐ | $\lambda = 3.17890, (X) = \begin{pmatrix} 1.231926 \\ 1.421389 \\ 1.0 \end{pmatrix}$ |
| ☐ | $\lambda = 3.17890, (X) = \begin{pmatrix} 1.0 \\ 1.421389 \\ 1.231926 \end{pmatrix}$ |
| ☒ | $\lambda = 3.17890, (X) = \begin{pmatrix} 1.0 \\ 1.231926 \\ 1.421389 \end{pmatrix}$ |





Question # 16 of 30 (Start time: 04:15:23 PM, 28 June 2021)

Which of the following system of equation is diagonally dominant?

Select the correct option



☐	$\begin{aligned}8x + y - z &= 5 \\x + 2y + 9z &= 3 \\2x + 13y + 5z &= 7\end{aligned}$
☐	$\begin{aligned}x + 2y + 9z &= 3 \\8x + y - z &= 5 \\2x + 13y + 5z &= 7\end{aligned}$
☐	$\begin{aligned}2x + 13y + 5z &= 7 \\8x + y - z &= 5 \\x + 2y + 9z &= 3\end{aligned}$
☒	$\begin{aligned}8x + y - z &= 5 \\2x + 13y + 5z &= 7 \\x + 2y + 9z &= 3\end{aligned}$

[Click to Save Answer & Mark](#)



Question # 17 of 30 (Start time: 04:15:54 PM, 28 June 2021)

The linear equation: $0x+0y=2$ has ----- solution/solutions.

Select the correct option

☐	finite many
☒	no solution
☐	unique
☐	infinite many

5/1/2021





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Question # 18 of 30 (Start time: 04:16:21 PM, 28 June 2021)

Which of the following is equivalent form of the system of equations in matrix form; $AX=B$?

Select the correct option

- | | |
|----------------------------------|-------------------------------|
| ☐ | $BX = A$ |
| ☐ | $XA = B$ |
| ☐ | $X = B(\text{Inverse of } A)$ |
| ☒ | $X = (\text{Inverse of } A)B$ |

Click





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Question # 19 of 30 (Start time: 04:16:51 PM, 28 June 2021)

In Jacobi's Method, We assume that theelements does not vanish.

Select the correct option

☒	Diagonal
☐	Row
☐	Column
☐	Off-diagonal





Question # 20 of 30 (Start time: 04:17:15 PM, 28 June 2021)

While using Relaxation method, which of the following is the Residuals for 1st iteration on the system; $2x+3y = 1$, $3x +2y =4$?

Select the correct option

- | | |
|----------------------------------|--------|
| ☐ | (-2,3) |
| ☐ | (3,-2) |
| ☒ | (1,4) |
| ☐ | (2,3) |





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Question # 21 of 30 (Start time: 04:17:44 PM, 28 June 2021)

There are ... types of error.

Select the correct option

☒	3
☐	5
☐	2
☐	4





Question # 22 of 30 (Start time: 04:18:09 PM, 28 June 2021)

Total Marks: 1

Which of the following is a forward difference table for the given values of x and y ?

x	0.1	0.7	1.3
y	0.003	0.248	0.697

Select the correct option

[Reload Math Equations](#)

☐	<table><tr><td>x</td><td>y</td><td>Δy</td><td>$\Delta^2 y$</td></tr><tr><td>0.1</td><td>0.003</td><td>0.251</td><td>0.1196</td></tr><tr><td>0.7</td><td>0.248</td><td>0.945</td><td></td></tr><tr><td>1.3</td><td>0.697</td><td></td><td></td></tr></table>	x	y	Δy	$\Delta^2 y$	0.1	0.003	0.251	0.1196	0.7	0.248	0.945		1.3	0.697		
x	y	Δy	$\Delta^2 y$														
0.1	0.003	0.251	0.1196														
0.7	0.248	0.945															
1.3	0.697																
☒	<table><tr><td>x</td><td>y</td><td>Δy</td><td>$\Delta^2 y$</td></tr><tr><td>0.1</td><td>0.003</td><td>0.245</td><td>0.204</td></tr><tr><td>0.7</td><td>0.248</td><td>0.449</td><td></td></tr><tr><td>1.3</td><td>0.697</td><td></td><td></td></tr></table>	x	y	Δy	$\Delta^2 y$	0.1	0.003	0.245	0.204	0.7	0.248	0.449		1.3	0.697		
x	y	Δy	$\Delta^2 y$														
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x	y	Δy	$\Delta^2 y$														
0.1	0.003	0.449	0.204														
0.7	0.248	0.245															
1.3	0.697																





Question # 23 of 30 (Start time: 04:18:50 PM, 28 June 2021)

The convergence of which of the following method is sensitive to starting value

Select the correct option

☐	False position
☒	Newton-Raphson method
☐	Gauss Seidal method
☐	Secant method

Click to Save Ans





Question # 24 of 30 (Start time: 04:19:17 PM, 28 June 2021)

If the Relaxation method is applied on the system; $2x+3y = 1$, $3x + 2y = -4$, then largest residual in 1st iteration will reduce to -----.

Select the correct option

☐	-1
☐	-1
☒	zero
☐	4

Question # 25 of 30 (Start time: 04:19:52 PM, 28 June 2021)

While using the relaxation method for finding the solution of the below given system, which of the following increment will be introduced?

$$\begin{aligned}6x_1 - 2x_2 + 3x_3 &= 1 \\ -2x_1 + 7x_2 + 2x_3 &= 5 \\ x_1 + x_2 - 5x_3 &= -13\end{aligned}$$

Select the correct option

[Reload Math](#)

☐	$dx_1 = \frac{R_1}{a_{11}}$
☐	$dx_2 = \frac{R_2}{a_{22}}$
☐	$dx_1 = \frac{R_1}{a_{22}}$
☒	$dx_3 = \frac{R_3}{a_{33}}$

[Click to Save Answer & Move to Next](#)



Question # 26 of 30 (Start time: 04:20:22 PM, 28 June 2021)

While solving a system of three linear equations in three variables by Jacobi's method, the first approximate solution is NECESSARY to be taken as (0,0,0).

Select the correct option

☐	False
☐	True

[Click to Save Answer & Move to Next](#)

Question # 27 of 30 (Start time: 04:20:45 PM, 28 June 2021)

Total Marks:

If $A = \begin{bmatrix} 2 & 0 & 0 \\ 3 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}$ then by using Gaussian Elimination method the value of A^{-1} will be

Select the correct option

[Reload Math Equations](#)

☐	$\begin{bmatrix} -\frac{1}{2} & 0 & 0 \\ 2 & -\frac{3}{2} & \frac{1}{2} \\ -\frac{5}{6} & \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$
☒	$\begin{bmatrix} \frac{1}{2} & 0 & 0 \\ -2 & \frac{3}{2} & -\frac{1}{2} \\ \frac{5}{6} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix}$
☐	$\begin{bmatrix} \frac{5}{6} & -\frac{2}{3} & \frac{1}{3} \\ -2 & \frac{3}{2} & -\frac{1}{2} \\ \frac{1}{2} & 0 & 0 \end{bmatrix}$
☐	$\begin{bmatrix} -2 & \frac{3}{2} & -\frac{1}{2} \\ \frac{1}{2} & 0 & 0 \\ \frac{5}{6} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix}$





Question # 28 of 30 (Start time: 04:21:21 PM, 28 June 2021)

Non-convergence in Newton-Raphson method can occur if the initial value is selected such that the derivative of the function equals to

Select the correct option

☐	Negative infinity
☒	Zero
☐	One
☐	Infinity

Click to Save Answer

MTH603:Grand Quiz

Quiz Start Time: 04:08 PM, 28 June 2021

Question # 29 of 30 (Start time: 04:21:49 PM, 28 June 2021)

Total Marks: 1

While solving a system of linear equations by Gauss Jordan Method, after all the elementary row operations if there lefts also zeros on the main diagonal then which of the is true about the system?

Select the correct option

☐	System may have unique solutions
☐	System may have infinite many solutions
☒	System has no solution
☐	System may have multiple numbers of finite solutions

Click to Save Answer & Move to Next Question





MTH603:Grand Quiz

Question # 30 of 30 (Start time: 04:22:16 PM, 28 June 2021)

For differences methods we require the initial approximation.

Select the correct option

☐	TRUE
☒	FALSE

Best short lectures by Tutor Sadiq Ahmed

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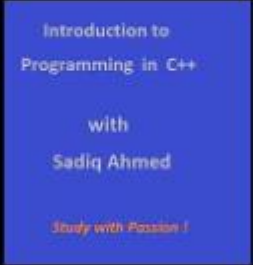
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- 4) Topic wise Notes
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- 6) MCQs Solved
- 7) Solved Past and Current paper
- 8) Solved Graded Activities (Assignment Quiz GDB)

Activity
Detailed
Information

Project Enrolment Complete training with Live classes on Zoom

- 1) SRS (Software Requirements Specification)
- 2) DD (Design Document)
- 3) Test phase
 - a) Test phase viva (Live Classes for viva preparation)
- 4) Final Deliverable
 - a) Final coding
 - b) Final presentation
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